

ELECTIVE-II

UCH701: CATALYTIC PROCESSES

L	T	P	Cr
3	1	0	3.5

Course Objective:

To gain the knowledge of catalyst characteristics, mechanism of catalytic reactions, and design of catalytic reactors.

Introduction: Catalysis and catalysts – homogeneous & heterogeneous, Classification of catalytic reactions and catalysts, Commercial chemical catalysts, Steps in catalytic reactions.

Preparation and Properties of Catalysts: Methods of catalyst preparation, Physical properties of catalyst – surface area, pore volume, pore size distribution, solid density, particle density, bulk density, void volume, Catalyst promoters & inhibitors, Catalyst accelerators & poisons.

Adsorption and Catalytic Reactions: Adsorption isotherms, Surface reaction, Single site and dual site mechanism, Desorption, Catalyst deactivation, Pore structure and surface area estimation and their significance.

External Transport Processes: Fluid particle mass and heat transfer, Mass transfer-limited reactions in packed beds, Nonisothermal behavior of packed-bed reactors, Staged packed-bed reactors for approaching optimum temperature progression, Stable operating conditions in reactors and hot spot formation, Effect of external transport processes on selectivity under non-isothermal conditions.

Diffusion and Reaction in Porous Catalysts: Intra-pellet mass transfer and diffusion in cylindrical and spherical porous catalyst particles, Thiele modulus, Diffusion controlled and surface reaction-controlled kinetics, Effectiveness factor for catalysts, Effects of heat transfer – temperature gradients across fluid-solid film and across catalyst pellet, Fluidized bed reactors, Three phase reactors – slurry and trickle bed reactors.

Generalized Design: Design of catalytic reactors under adiabatic and non-adiabatic conditions, Design of industrial fixed-bed, fluidized-bed and slurry reactors.

Course Learning Outcomes (CLO):

Upon completion of the course, students will be able to:

1. identify mechanism for various catalytic reactions.
2. select suitable characterization technique for different catalysts.
3. solve problems related to external heat and mass transfer effects in heterogeneous catalysis.
4. calculate the effectiveness of a porous catalyst.
5. design different types of reactors for catalytic reactions.

Text Books:

1. Smith, J.M., *Chemical Engineering Kinetics*, McGraw-Hill (1981).
2. Fogler, H.S., *Elements of Chemical Reaction Engineering*, Prentice-Hall India (2009).

Reference Books:

1. Denbigh, K.G., and Turner, J.C.R., *Chemical Reactor Theory: An Introduction*, Cambridge University Press (1984).
2. Carberry, J.J., *Chemical and Catalytic Reaction Engineering*, McGraw-Hill, (2001).
3. Levenspiel, O., *Chemical Reaction Engineering*, John Wiley (2006).

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	45
3	Sessional (may include Quizzes/Assignments/Tutorials evaluations)	25